# Game Assistant
# tech stack proposal

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Pipeline

Voice input

Text input

LLM

Text output

Speech output

# Voice input → Text input

- Use pyaudio to capture voice
- Use "faster-whisper" or "whisper-tiny" for Speech to Text

# Text input → LLM

- Combine the user prompt with the current game information and the most important Wiki information
  - Minecraft game information using our own Fabric mod (that provides info in JSON)
  - Add the most relevant Wiki pages to the prompt as well (use RAG = "Retrieval-augmented generation")
- Provide text to LLM
- LLM can be run locally using Ollama
- Choices for lightweight LLMs:
  - Meta Llama 3.2
  - Qwen 3

# LLM → Text output → Speech

- Take the text output of LLM and turn it into speech using "Piper"
- Additional feature: give user option to choose different voice.

# General

- UI: provide a minimal UI to control speech input and view if the connection to the game (via the mod) works or not.
- Don’t retrain using wikis; use RAG